# Alex Doe

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Brookfield, ST | Willing to relocate

## Experience

**Embedded Firmware Intern, Example Motor Controls**, Fairview, ST — Summer 2025

- Added a 200 ms soft-start ramp to the STM32 PWM firmware for a 24 V brushless drive, which lowered startup current spikes by 35% across 50 starts measured with a current probe
- Cut bring-up from 2 days to 4 hours with a Python script that tests 40 channels over UART
- Fixed a CAN bus bug that dropped 1 message in 200 by measuring bit timing on an oscilloscope and moving the sample point to 87.5%, then ran a 12-hour soak test with no dropped messages

**Test Engineering Intern, Example Power Systems**, Brookfield, ST — Summer 2024

- Automated a 20-point load sweep in LabVIEW with a programmable load and a power analyzer, so an efficiency test on the 500 W supply took 25 minutes instead of the 3 hours it took by hand
- Found a 15 °C hot spot that the derating curve missed on a 12 V module by logging 8 thermocouples through a 48-hour full-load test, and the design team enlarged the heatsink on the next revision

## Projects

**Senior Design: Solar Charge Controller** (KiCad, STM32, C, LTspice) — Fall 2025 – Spring 2026

- Wrote perturb-and-observe MPPT firmware in C for an STM32 that updates every 10 ms and harvested 18% more energy than a fixed-voltage controller in tests on a solar array simulator
- Laid out a 100 W buck stage on a 2-layer KiCad board that stays under 70 °C at full load
- Sized the output filter in LTspice and measured 42 mV of ripple on the oscilloscope

**Digital Audio Filter on an FPGA** (Verilog, Vivado, MATLAB, Basys 3) — Spring 2025

- Built a 32-tap FIR low-pass filter in Verilog on a Basys 3 FPGA board running at 48 kHz, with 16-bit fixed-point coefficients that matched the MATLAB design within 0.3 dB in Vivado simulation

## Education

**B.S. Electrical Engineering**, GPA 3.71/4.00 — Spring 2026
Example State University
**Relevant coursework:** Embedded Systems, Power Electronics, Modern Control Design
**Activities:** Eta Kappa Nu

## Skills

**Software:** C, Python, MATLAB, Verilog, KiCad, LTspice, LabVIEW
**Hardware/Lab:** Oscilloscope, Current probe, Programmable load, Power analyzer, STM32, FPGA, CAN bus
**Certifications:** Engineer in Training (EIT), IPC-A-610 Certified IPC Specialist